

Hackathon Problem Statement:

SentinelAI

1. Project Title

SentinelAI: Multi-Modal Real-Time Sepsis Forecasting with Explainable Clinical Decision Support

2. Executive Summary

Develop a real-time predictive analytics engine designed to forecast the **48-hour risk of Sepsis onset** in hospitalized adult patients. The solution must break down clinical data silos by integrating **Electronic Health Records (EHR)**, **high-frequency wearable sensor data**, and **Social Determinants of Health (SDOH)** to provide a holistic, proactive risk profile.

3. The Problem

Sepsis is a leading cause of hospital mortality, where every hour of delayed treatment increases the risk of death by nearly 8%. Current detection systems often rely on "late-stage" physiological triggers (like the qSOFA score), leading to reactive rather than proactive care. Furthermore, most AI models in healthcare are "black boxes," lacking the transparency needed for clinicians to trust their outputs in high-stakes environments.

4. The Challenge

Participants must build a solution that satisfies the following four pillars:

I. Multi-Modal Data Integration

- **Static/Historical:** Integration of EHR data (demographics, lab results, medical history).
- **Dynamic/Streaming:** Processing of high-frequency vitals (Heart Rate, SpO2, Respiratory Rate) from wearable sensors.
- **Contextual:** Inclusion of SDOH factors (e.g., Social Vulnerability Index, housing stability, or proximity to care) to identify at-risk populations more accurately.

II. Explainable AI (XAI) for Clinical Trust

- The model must not only provide a risk score but also **visualize the "Why."**
- Use interpretability frameworks (e.g., SHAP, LIME) to highlight the top 3-5 physiological or social drivers behind a high-risk alert.

III. Low-Latency Performance

- To support bedside monitoring and rapid response teams, the system must achieve an

end-to-end inference latency of **<500ms**.

IV. Workflow Integration & Interoperability

- The solution must be **FHIR-compliant** (Fast Healthcare Interoperability Resources).
- Outputs should be formatted as FHIR RiskAssessment or Observation resources to allow seamless embedding into modern EHR systems like Epic or Cerner.

5. SWOT Analysis

Strengths	Weaknesses
- High technical barrier to entry (Multi-modal data). - Explainability (XAI) fosters high clinical adoption. - Real-time processing via low-latency architecture.	- High dependency on data quality from wearable sensors. - Initial integration complexity with legacy hospital systems.
Opportunities	Threats
- Expansion into other acute conditions (Heart Failure, AKI). - Partnership with wearable device manufacturers. - Government incentives for improving patient safety metrics.	- Stringent healthcare data privacy regulations (HIPAA/GDPR). - Potential "alarm fatigue" for clinicians if not properly tuned.

6. Revenue Model (Sustainability & Monetization)

To ensure long-term viability, SentinelAI adopts a multi-tiered commercial strategy:

1. **SaaS Subscription (B2B):** Annual licensing fee per hospital bed or per department (ICU/Emergency Ward) for integration with their EHR.
2. **Implementation Services:** One-time integration fee for mapping local hospital data structures to FHIR-compliant formats.
3. **Performance-Based Incentives:** A "Value-Based Care" model where the hospital shares a percentage of the savings generated by reduced ICU stays and decreased mortality rates.
4. **Data Insights (Aggregated & De-identified):** Providing anonymized, high-level population

health trends to public health agencies and research institutions (subject to strict ethical guidelines).

7. Technical Deliverables

1. **Predictive Model:** A machine learning pipeline optimized for 48-hour sepsis forecasting.
2. **XAI Dashboard:** A UI/UX prototype showing the risk score alongside feature importance.
3. **FHIR API:** A documented API endpoint returning FHIR-standardized risk objects.
4. **Actionable Insights:** A recommended "Next Best Action" (e.g., "Order STAT Lactate").

8. Success Metrics

- **Clinical Accuracy:** Measured by AUROC and AUPRC for sepsis prediction.
- **Technical Efficiency:** Latency benchmarks for data processing and model inference.
- **Trust Score:** Qualitative assessment of the explainability visualization.
- **Interoperability:** Validation of the FHIR-standardized JSON output.